# Introduction to Research and Identification of Research Problem: A Structured Academic Framework for Computer Science Students

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***Abstract*—Research is the systematic process by which new knowledge is created, existing knowledge is tested, and practical problems are resolved. For Computer Science (CS) students, understanding how to identify, formulate, and articulate a research problem is a foundational academic competency. This document provides a comprehensive academic treatment of the research process, classifications of research, and step-by-step formulation of a research problem, research gap, research question, and research objective. A worked CS example — skin cancer detection using deep learning — is used throughout to ground abstract concepts in practical application. IEEE citation conventions are used throughout.**

***Index Terms*—research methodology, research problem, research gap, research question, research objective, problem statement, literature review, computer science research, deep learning, classification**

## I. Introduction

Every technological advancement begins with a research problem. Before an experiment is designed, before a dataset is collected, and before a single line of code is written, a researcher must ask: *What is the problem? Why does it matter? What has already been attempted? What remains unresolved?*

These questions form the logical foundation of the research process. Students entering the research domain frequently encounter difficulty at this first stage: identifying and formulating a research problem [1].

A well-formulated research problem is fundamental to achieving practical relevance in research; studies show a persistent gap between academic contributions and industrial needs, often attributed to poorly formulated research problems and the lack of structured approaches to evaluate them [2].

This document addresses each element of the research problem formulation process in a structured, academically rigorous manner.

## II. Understanding Research

### A. Definition

Research is the systematic process of investigating questions, testing hypotheses, and generating new knowledge, involving collecting, analysing, and interpreting data to answer a specific question or solve a defined problem [3].

### B. Characteristics of Good Research

Good research is systematic, rigorous, empirical, replicable, valid, reliable, objective, and peer-reviewed [3].

### C. Domains of Research

TABLE I
Research Domains and Examples

| Domain | Focus | CS Example |
|---|---|---|
| Scientific | Natural laws | Thermodynamics |
| Academic | Knowledge advancement | Complexity theory |
| Engineering | System design | Network routing |
| CS Research | Algorithms, AI | Medical image DL |
| Applied | Practical problems | Fraud detection |
| Basic | Foundational theory | P vs NP |

## III. Types of Research

It is important to note that research classifications are **not mutually exclusive**. A single study may simultaneously be quantitative, experimental, and applied [1].

### A. Basic / Fundamental Research

Research conducted to advance theoretical knowledge without immediate practical application. CS Example: Formal verification of algorithm correctness; P vs NP.

### B. Applied Research

Research conducted to solve a specific, practical problem using existing knowledge [3]. CS Example: Real-time object detection for autonomous vehicles.

### C. Quantitative Research

Quantitative research refers to the systematic empirical investigation of phenomena via statistical, mathematical, or computational techniques [3]. CS Example: Measuring classification accuracy of machine learning algorithms across a benchmark dataset.

### D. Qualitative Research

Qualitative research gathers data about lived experiences and meanings, investigating the why and how of phenomena [1]. CS Example: Investigating why developers resist agile adoption via structured interviews.

### E. Mixed-Methods Research

Mixed-methods research uses strategies from both qualitative and quantitative processes to provide greater understanding [1]. CS Example: Measuring task completion rates quantitatively, then interviewing users about usability failures qualitatively.

### F. Experimental Research

Experimental research designs phenomena under strictly controlled conditions, manipulating independent variables and measuring dependent variables to establish causation [1]. CS Example: Comparing classification accuracy of ResNet-50 vs. VGG-16 under identical conditions.

### G. Exploratory Research

Preliminary investigation of an understudied area to generate hypotheses [3]. CS Example: Exploring quantum computing for cryptographic applications.

### H. Descriptive Research

Systematic description of characteristics of a phenomenon or population [3]. CS Example: Surveying programming language distribution in open-source repositories.

### I. Explanatory Research

Research that identifies causal mechanisms explaining observed phenomena [3]. CS Example: Why transformer models outperform recurrent networks on long-sequence NLP tasks.

### J. Correlational Research

Examines relationships between variables without establishing causation [1]. CS Example: Correlation between dataset size and model accuracy.

### K. Case Study Research

In-depth investigation of a single instance in its real-world context [4]. CS Example: Software development process at a DevOps organization.

### L. Survey Research

Systematic data collection from a population using standardized instruments [3]. CS Example: Surveying engineers about productivity with AI coding assistants.

### M. Action Research

Iterative practitioner research for solving immediate practical problems [5]. CS Example: Iteratively refining sprint planning using reflection cycles.

### N. Empirical Research

Research based on observation, investigation, or experiment rather than on abstract reasoning or speculation [6]. CS Example: Training and evaluating a sentiment analysis model on a Twitter dataset.

## IV. The Research Process

The research process is a sequential yet iterative chain of activities that transforms a vague area of interest into a definitive, evidence-based contribution to knowledge [1].

Each stage is described below:

1) **Research Area:** The broadest domain of inquiry. Example: Artificial Intelligence.
2) **Research Topic:** A specific subject within the area. Example: Deep Learning for Medical Image Classification.
3) **Literature Search:** Systematic identification of relevant published work using IEEE Xplore, ACM DL, Google Scholar, and Semantic Scholar.
4) **Literature Review:** Critical synthesis of existing knowledge: what has been studied, what methods have been used, what are the contradictions, and what remains open [7].
5) **Research Problem:** An unresolved issue, contradiction, or gap in existing knowledge that warrants investigation.
6) **Research Gap:** The specific missing knowledge identified from the literature review.
7) **Research Question:** A precise, testable question derived from the gap.
8) **Research Objectives:** Measurable actions to be undertaken to answer the research question.
9) **Research Methodology:** The research design, data collection strategy, analytical approach, and evaluation criteria.
10) **Data Collection / Dataset:** Selecting or constructing a dataset; preprocessing; splitting; documentation.
11) **Experiment / Analysis:** Execution of the methodology under documented, reproducible conditions.
12) **Results:** Objective outputs of the experiment reported without interpretation.
13) **Discussion:** Interpretation of results in the context of the research question and existing literature.
14) **Conclusion:** Synthesis of findings; whether objectives were achieved; contribution statement.
15) **Future Work:** Limitations of the current study; directions for subsequent investigation.

## V. Identification of Research Problems

### A. Sources of Research Problems

Research problems arise from multiple sources [1], [3]:

1) Limitations of existing studies
2) Contradictory findings in the literature
3) Unsolved technical problems
4) Practical operational needs lacking technical solutions
5) Theoretical gaps: frameworks not tested in a given context
6) Methodological limitations of prior work
7) Dataset limitations: small, biased, or domain-specific data
8) Emerging technologies not yet applied to established problems

9) Replication studies: results not independently reproduced
10) Extension of prior research to new domains or contexts

### B. How to Read a Paper to Find a Research Problem

When reading a paper, the following linguistic signals often indicate candidate research problems. Each requires critical evaluation — not every instance constitutes a researchable gap.

TABLE II
SIGNAL WORDS FOR RESEARCH GAP IDENTIFICATION

| Signal Word | What It May Indicate |
|---|---|
| "However" | A limitation or contradiction being introduced |
| "Despite" | Unresolved tension between expectation and reality |
| "Limitation" | Acknowledged boundary of the study |
| "Future Work" | Open problems identified by authors |
| "Challenge" | Known obstacles not yet overcome |
| "Open Problem" | Explicitly unresolved questions |
| "Gap" | Directly stated absence of knowledge |
| "Remains" | Persisting unresolved state of affairs |
| "Generalization" | Whether results transfer beyond study context |
| "Computational Cost" | Practical limitation of a viable approach |

**Critical Warning:** The presence of "however" does not automatically indicate a research gap. The candidate gap must be: (a) unresolved in the literature, (b) consequential, (c) researchable, and (d) feasible within the researcher's constraints.

### C. Problem Statement Structure

A well-constructed problem statement follows this structure:

1) **Context** — Establish the domain and significance
2) **Existing situation** — What exists and what has been done
3) **Existing limitation** — What is insufficient
4) **Consequence** — The consequence of this limitation
5) **Research gap** — The specific missing knowledge
6) **Need for investigation** — Why this must be studied

## VI. RESEARCH GAPS

Research gaps refer to unanswered questions in the existing body of knowledge, either due to a lack of studies or inconclusive results [8]. They are essential starting points and motivation in scientific research [8].

Types of research gaps include knowledge gaps, methodological gaps, empirical gaps, theoretical gaps, data gaps, population/context gaps, contradictory evidence, technology gaps, evaluation gaps, and generalization gaps [9].

## VII. FORMULATION OF RESEARCH QUESTIONS

### A. Characteristics of a Strong Research Question

A strong research question must be [1]:

- **Clear** — unambiguous and well-defined
- **Specific** — bounded in scope
- **Researchable** — answerable through data collection and analysis
- **Feasible** — achievable within time and resource constraints
- **Connected to a gap** — derived from literature
- **Measurable** — where empirical, answerable with data
- **Methodologically aligned** — the method can answer it

### B. Forms of Research Questions

TABLE III
FORMS OF RESEARCH QUESTIONS WITH CS EXAMPLES

| Form | Purpose | CS Example |
|---|---|---|
| Descriptive | Describe phenomenon | What DL architectures are used for medical imaging? |
| Comparative | Compare entities | ResNet-50 vs. EfficientNet-B4 on ISIC 2019? |
| Relational | Examine relationships | Correlation between dataset size and accuracy? |
| Causal | Establish cause-effect | Does focal loss reduce minority-class error? |
| Evaluative | Assess effectiveness | How effective is transfer learning under limited data? |
| Exploratory | Investigate new area | Vision transformers for real-time medical diagnostics? |

### C. Bad vs. Good Research Questions

TABLE IV
BAD VS. GOOD RESEARCH QUESTIONS

| Bad RQ | Good RQ |
|---|---|
| "What is artificial intelligence?" | "How does transfer learning affect multi-class image classification performance under limited training data?" |
| "Which is the best machine learning algorithm?" | "How does Random Forest compare with SVM for network intrusion detection on NSL-KDD under identical preprocessing conditions?" |
| "How can we improve deep learning?" | "Does focal loss improve minority-class recall in a seven-class skin lesion classifier on ISIC 2019 compared to cross-entropy loss?" |

## VIII. FORMULATION OF RESEARCH OBJECTIVES

Research objectives must use measurable action verbs: investigate, evaluate, compare, analyze, develop, design, implement, assess, determine, optimize, validate [1].

Vague verbs such as "understand," "know," and "learn" should be avoided unless precisely operationalized.

## IX. EXAMPLE: COMPUTER SCIENCE RESEARCH PROBLEM

**Research Area:** Artificial Intelligence / Computer Vision

**Research Topic:** Skin Cancer Detection Using Deep Learning

**Research Problem:** Existing skin lesion classification models achieve high overall accuracy but exhibit reduced sensitivity on minority lesion classes due to class imbalance.

TABLE V
RESEARCH QUESTION TO OBJECTIVE MAPPING

| Research Question | Research Objective |
|---|---|
| How does architecture choice affect classification performance? | To evaluate ResNet-50, VGG-16, and EfficientNet-B4 under identical conditions using macro-F1 and per-class recall on ISIC 2019. |
| Does focal loss improve minority-class detection? | To compare per-class recall under focal loss vs. cross-entropy under identical training conditions. |

Cross-dataset generalization of class-balancing strategies has not been adequately evaluated in the literature.

**Research Gap:** The combined effect of data augmentation and class-weighted loss on minority-class recall has not been systematically evaluated across multiple publicly available dermatological datasets.

**Research Questions:**

RQ1: Does the combination of data augmentation and class-weighted loss improve minority-class recall in EfficientNet-B4 trained on ISIC 2019 compared to a cross-entropy baseline?

RQ2: Do these performance improvements generalize from ISIC 2019 to HAM10000 without additional fine-tuning?

**Research Objectives:**

O1: To implement and evaluate five augmentation strategies combined with class-weighted loss on EfficientNet-B4 trained on ISIC 2019.

O2: To assess per-class recall, macro-F1, and overall accuracy for each experimental condition.

O3: To evaluate generalization on HAM10000 without retraining.

O4: To measure and compare training cost across conditions.

**Dataset:** ISIC 2019 [10]; HAM10000 [11]

**Methodology:** Controlled quantitative experimental design; EfficientNet-B4; four experimental conditions; evaluation via macro-F1, per-class recall, AUC-ROC.

## X. DISCUSSION

The process of research problem formulation is a systematic, evidence-driven process beginning with thorough engagement with the existing literature. The importance of the research problem lies in the fact that the entire research effort begins with articulation and formulation of a researchable problem [3].

Three critical errors frequently committed by novice researchers are: (1) confusing a topic for a problem; (2) asserting a research gap without evidence; (3) formulating a research question that cannot be answered by the chosen methodology.

A well-identified gap transforms research from an academic exercise into a contribution that advances knowledge and improves practice [12].

## XI. CONCLUSION

This document has provided a comprehensive IEEE-organized academic treatment of research fundamentals, types of research, the research process, and systematic formulation of a research problem. The logical chain:

*Research Area* → *Topic* → *Literature* → *Problem* → *Gap* → *Research Question* → *Objective* → *Methodology* → *Evidence* → *Result* → *Conclusion*

must be maintained at every stage. Every element must be logically connected and evidentially supported. The worked skin cancer detection example demonstrates that complex research problems can be systematically decomposed into precise, measurable, feasible research questions when this process is followed rigorously.

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